Machine Edition

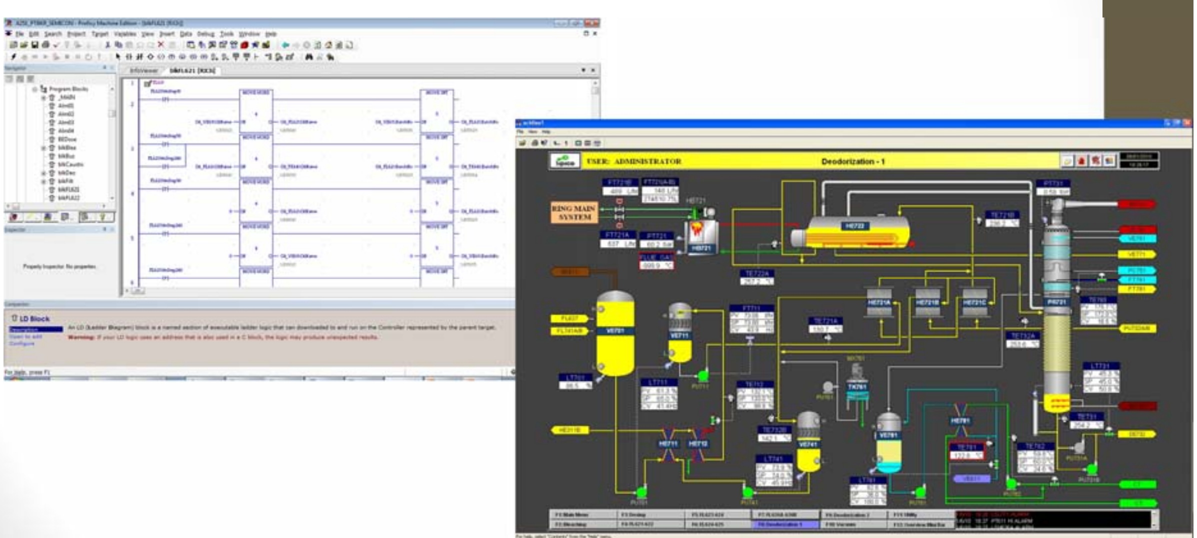
M. Yazid

06 Agustus 2011

WILMAR GROUP

GE SCADA

- **Programming** : Logic Master, Versapro, Machine Edition
- **Human machine Interface** : Cimplicity, Citect, Wonderware, dll



PT. WILMAR NABATI INDONESIA - GRESIK

WILMAR GROUP

Machine Edition

- Contact
- Coil
- Math
- Data Move
- Timer
- Counter
- Relational

PT. WILMAR NABATI INDONESIA - GRESIK

WILMAR GROUP

Address Register

1. Field/Hardwired :
   - Digital Input : %I
   - Digital Output : %Q
   - Analog Input : %AI
   - Analog Output : %AQ.
2. Memory :
   - %M
   - %G
   - %T
   - %SA/%SB
   - %R

WILMAR GROUP

Contact

Type of Contact	Display	Contact Passes Power to Right
Normally Open	—| |—	When reference is ON.
Normally Closed	—|/|—	When reference is OFF.
Continuation Contact	<+>———	If the preceding continuation coil is set ON.

WILMAR GROUP

Coil

No.	Type of Coil	Display	Power to Coil	Result
1	Normally	—()—	ON	Sets reference ON.
	Open		OFF	Sets reference OFF.
2	Negated	—(/)—	ON	Sets reference OFF.
			OFF	Sets reference ON.
3	Retentive	—(M)—	ON	Sets reference ON, retentive.
			OFF	Sets reference OFF, retentive.
4	Negated	—(/M)—	ON	Sets reference OFF, retentive.
	Retentive		OFF	Sets reference ON, retentive.
5	Positive	—(↑)—	OFF→ON	If reference is OFF, sets it ON for one sweep.
	Transition			
6	Negative	—(↓)—	ON←OFF	If reference is OFF, sets it ON for one sweep.
	Transition			
7	SET	—(S)—	ON	Sets reference ON until reset OFF by —(R)—.
			OFF	Does not change the coil state.
8	RESET	—(R)—	ON	Sets reference OFF until set ON by —(S)—.
			OFF	Does not change the coil state.
9	Retentive SET	—(SM)—	ON	Sets reference ON until reset OFF by —(RM)—,
				retentive.
			OFF	Does not change the coil state.
10	Retentive	—(RM)—	ON	Sets reference OFF until set ON by —(SM)—,
				retentive.
	RESET		OFF	Does not change the coil state.
11	Continuation	——<+>	ON	Sets next continuation contact ON.
	Coil		OFF	Sets next continuation contact OFF.

Timer and Counter

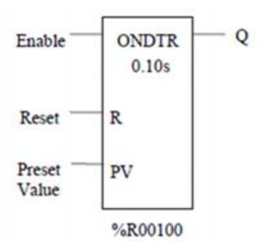
Abbreviation	Function
ONDTR	Retentive On-Delay Timer
TMR	Simple On-Delay Timer
OFDT	Off-Delay Timer
UPCTR	Up Counter
DNCTR	Down Counter

PT. WILMAR NABATI INDONESIA - GRESIK

WILMAR GROUP

Timer and Counter

On Delay Timer

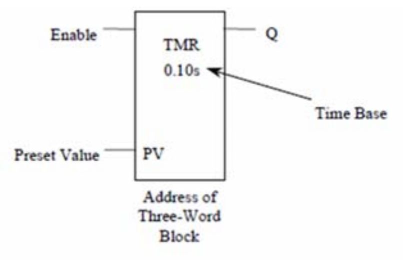


PT. WILMAR NABATI INDONESIA - GRESIK

WILMAR GROUP

Timer and Counter

Timer

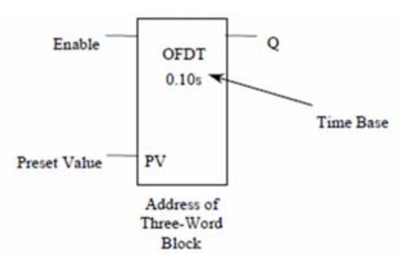


PT. WILMAR NABATI INDONESIA - GRESIK

WILMAR GROUP

Timer and Counter

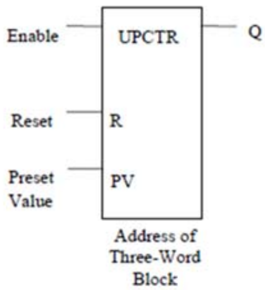
Off Delay Timer



Timer and Counter

WILMAR GROUP

Up Counter

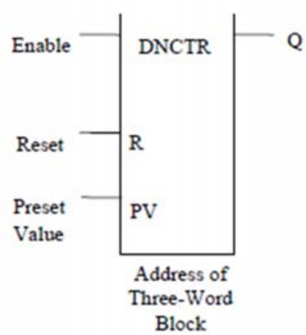


PT. WILMAR NABATI INDONESIA - GRESIK

Timer and Counter

WILMAR GROUP

Down Counter



Math

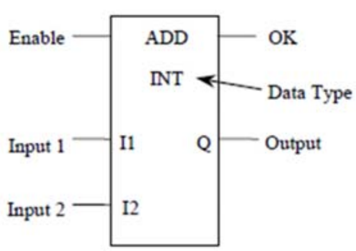
No.	Abbreviation	Function	Description
1	ADD	Addition	Add two numbers.
2	SUB	Subtraction	Subtract one number from another.
3	MUL	Multiplication	Multiply two numbers.
4	DIV	Division	Divide one number by another, yielding a quotient.
5	MOD	Modulo Division	Divide one number by another, yielding a remainder.
6	SQRT	Square Root	Find the square root of an integer or real value.
7	SIN, COS, TAN,	Trigonometric Functions †	Perform the appropriate function on the real
8	ASIN, ACOS,		value in input IN.
9	ATAN		
10	LOG, LN	Logarithmic/Exponential Function	Perform the appropriate function on the real
11	EXP, EXPT		value in input IN.
12	RAD, DEG	Radian Conversion †	Perform the appropriate function on the real
			value in input IN.

PT. WILMAR NABATI INDONESIA - GRESIK

WILMAR GROUP

Math

Add



Math

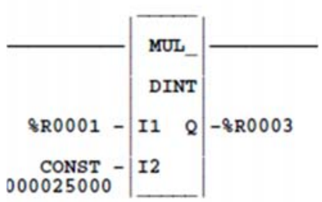
Data Type

- BOOL = 0 / 1
- INT = -32768 …. 32768
- DINT = -2147483648 … 2147483648
- REAL = $\pm 1.175494351 \times 10^{-38}$

PT. WILMAR NABATI INDONESIA - GRESIK
WILMAR GROUP

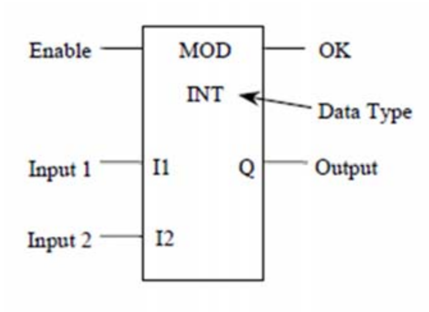
Math

Mul



Math

Mod



PT. WILMAR NABATI INDONESIA - GRESIK

Relational Function

Abbreviation	Function	Description
EQ	Equal	Test two numbers for equality.
NE	Not Equal	Test two numbers for non-equality.
GT	Greater Than	Test for one number greater than another.
GE	Greater Than or Equal	Test for one number greater than or equal to another.
LT	Less Than	Test for one number less than another.
LE	Less Than or Equal	Test for one number less than or equal to another.
RANGE	Range	Determine whether a number is within a specified range

WILMAR GROUP

Move

Abbreviation	Function	Description
MOVE	Move	Copy data as individual bits. The maximum length allowed is 256 words, except MOVE_BIT is 256 bits. Data can be moved into a different data type without prior conversion.